Apple HIG Cheat Sheet

Design Rules, Constraints & CSS Reference

CATEGORY	DESIGN RULE	APPLICATION GUIDELINE	MATH LIMIT	CSS BOILERPLATE
Typography	**Support fluid scaling**	Scale fonts using Dynamic Type while retaining visual hierarchy.	`Size ≥ 17pt`	`font: -apple-system, sans-serif;`
Color & Contrast	**Ensure accessible legibility**	Maintain contrast ratios across Light and Dark modes.	`CR ≥ 4.5:1`	`color: CanvasText;`
App Icons	**Design for scales**	Use vector graphics without baked-in drop shadows or text.	`r ≈ 0.225w`	`border-radius: 22.5%;`
Safe Areas	**Respect constraints**	Keep content out of the Dynamic Island and home indicator zones.	`Margin ≥ 16pt`	`padding-bottom: env(safe-area-inset-bottom);`
Buttons	**Provide clear touch targets**	Group primary actions and use ample padding around tap areas.	`Area ≥ 44×44pt`	`min-height: 44px; min-width: 44px;`
Context Menus	**Offer contextual shortcuts**	Reveal secondary actions without navigating away from the view.	`Offset ≥ 12pt`	`position: absolute; z-index: 50;`
Tab Bars	**Flatten app hierarchy**	Use for 3 to 5 mutually exclusive top-level destinations.	`Height = 49pt`	`position: fixed; bottom: 0;`
Sidebars	**Structure complex apps**	Use on iPadOS and macOS for fluid, multi-column navigation.	`W = 260-320pt`	`grid-template-columns: minmax(260px, 320px) 1fr;`
Text Fields	**Prevent layout shifts**	Align baselines and prevent auto-zoom on mobile devices.	`Height ≥ 44pt`	`font-size: 17px; appearance: none;`
Selection	**Immediate state toggles**	Apply toggles for binary states and sliders for ranges.	`r ≥ 14pt`	`border-radius: 999px; cursor: pointer;`
Scroll Views	**Predictable scrolling**	Avoid nesting scroll views that track the exact same axis.	`Offset ≥ 0`	`overscroll-behavior: contain;`

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Indicators	**Acknowledge system delays**	Show skeletons or spinners for tasks taking over a second.	`Delay > 1.0s`	`animation: spin 1s linear infinite;`
Alerts & Sheets	**Require explicit dismissal**	Use modal sheets for focused tasks to retain background context.	`Area ≈ 80%`	`position: fixed; inset: 10%;`
Activity Rings	**Cumulative progress**	Use closed rings strictly for daily goal-oriented metrics.	`Freq ≥ 60Hz`	`stroke-dasharray: 100; stroke-dashoffset: var(--v);`
Widgets	**Glanceable updates**	Refresh background content without draining system battery.	`Upd < 4/ hour`	`display: flex; flex-direction: column;`
Lists & Tables	**Scanning rhythm**	Standardize cell heights and inset separators to match text.	`Row ≥ 44pt`	`border-bottom: 1px solid var(--separator);`
Gestures	**Standard interactions**	Map swipes and pinches to expected navigation behaviors.	`V > 1000pt/s`	`touch-action: pan-y pinch-zoom;`
Charting Data	**Minimize cognitive load**	Limit the number of distinct data series and remove chart junk.	`Series ≤ 6`	`gap: 2px; display: flex;`